POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
STA. MESA MANILA
COLLEGE OF ENGINEERING

Self-Study Report (SSR)

Bachelor of Science in Civil Engineering

Criterion 2
Student Outcomes

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A LEADING COMPREHENSIVE POLYTECHNIC UNIVERSITY IN ASIA

QUALITY MANAGEMENT SYSTEM IS CERTIFIED ACCORDING TO ISO 9001
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WURI 250 OUT OF 400

2.1 Student Outcomes

The Bachelor of Science in Civil Engineering (BSCE) program of the Polytechnic University of the Philippines adopts and implements the Student Outcomes prescribed in CMO No. 92, s. 2017 (Attachment C2-10.2 – CHED Memorandum Order No 92 series of 2017 Policies, Standards and Guidelines for Bachelor of Science in Civil Engineering (BSCE) Program). Student Outcomes (SOs) describe the knowledge, skills, and professional attributes that students are expected to demonstrate upon graduation.

The twelve (12) SOs of the BSCE program are presented below. The formulation and adoption of these outcomes are guided by national and professional standards, including those prescribed by the CHED and the PTC. Together with the Program Education Objectives (PEOs), the SOs serve as the foundation for curriculum design, assessment, and continuous quality improvement of the program.

1. Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization to the solution of complex civil engineering problems **(Knowledge Competence)**.
2. Conduct investigations of complex civil engineering problems using research-based knowledge and research methods including design of experiments, analysis, and interpretation of data, and synthesis of information to provide valid conclusions **(Investigation)**.
3. Design solutions for complex civil engineering problems and design systems, components or processes that meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, cultural, and sustainability, in accordance with standards **(Design/Development of Solutions)**.
4. Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings **(Leadership and Teamwork)**.
5. Identify, formulate, research literature, and analyze complex civil engineering problems, reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences **(Problem Analysis)**.
6. Apply ethical principles and commit to professional ethics, and responsibilities and norms of engineering practice **(Ethics and Professionalism)**.
7. Communicate effectively on complex civil engineering activities with the engineering community and with society at large, such as being able to comprehend and write

effective reports and design documentation, make effective presentations, and give and receive clear instructions **(Communication)**.

8. Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex civil engineering problems in a global, economic, societal and environmental context **(Environment and Sustainability)**.
9. Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change **(Lifelong Learning)**.
10. Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues, and the consequent responsibilities relevant to professional engineering practice and solutions to complex civil engineering problems **(The Engineer and Society)**.
11. Create, select and apply appropriate techniques, skills, resources, and modern engineering and IT tools, including prediction and modelling, to complex civil engineering problems with an understanding of the limitations **(Modern Tool Usage)**.
12. Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one’s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments **(Project Management and Finance)**.

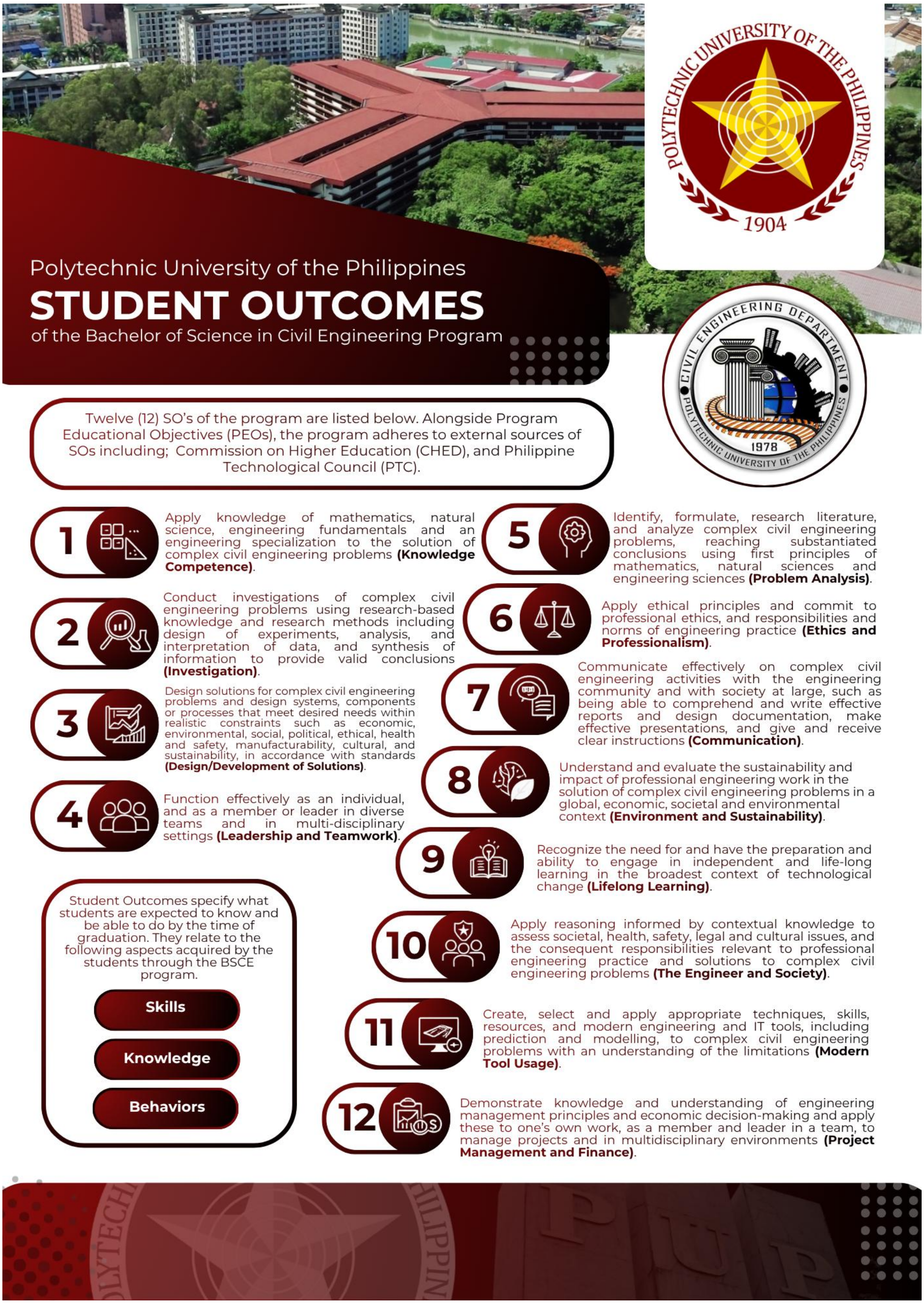

Figure 2-1. *Student Outcomes of the Bachelor of Science in Civil Engineering Program. The twelve SOs prescribed in CHED Memorandum Order No. 92, s. 2017 defines the knowledge, skills, and professional attributes expected of graduates. Guided by national and professional standards, these outcomes serve as the foundation for the program's curriculum design, assessment, and continuous quality improvement.*

The PUP SOs of BSCE are published on the school's website (https://www.pup.edu.ph/CE/BSCE), Outcomes-Based Teaching Learning Plan (OBTLP), and in the manual of quality systems management employees and students. This is also posted in the strategic locations inside the campus.

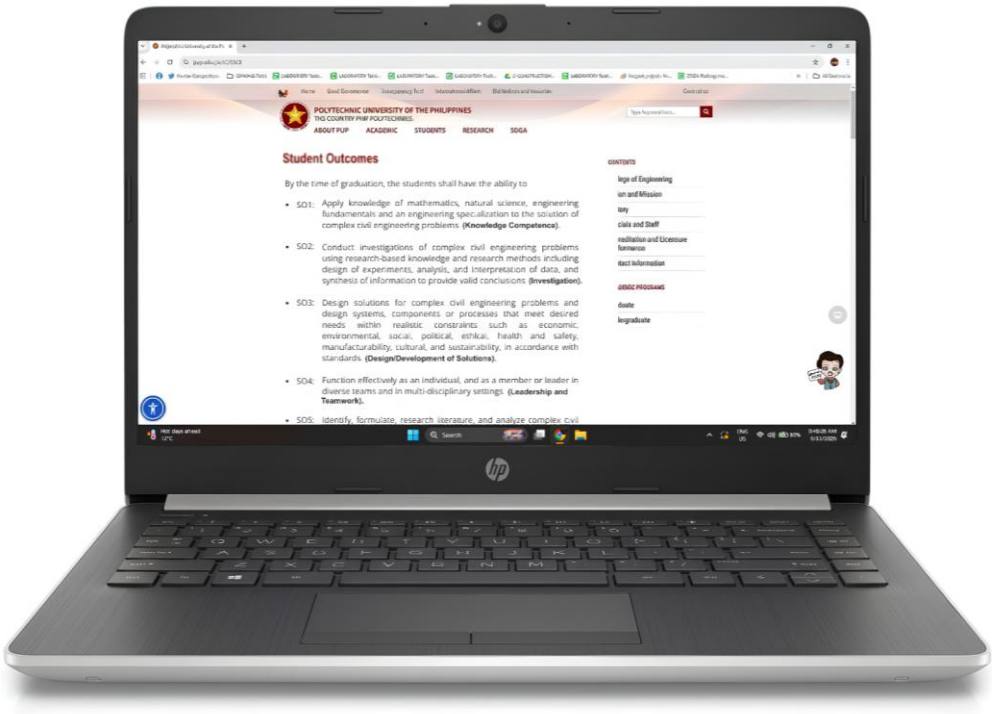

Figure 2-2. *Student Outcomes of the BSCE Program as published on the official PUP website. The framework is prominently accessible online to ensure transparency and continuous institutional alignment. It is also integrated into the Outcomes-Based Teaching Learning Plans, quality systems manuals, and strategic campus displays for stakeholder visibility.*

2.2 Relationship of Student Outcomes to Program Educational Objectives

Student Outcomes are necessary skills, knowledge, and behaviors students should acquire as they progress through the program, and they are intended to foster the achievement of Program Educational Objectives. It is important to relate each SO to the Program Educational Objectives.

The Program Educational Objectives are the following:

- **Demonstrating Engineering Profession:** Employed and/or practiced in the field of Civil Engineering such as but not limited to Construction Engineering and

Management, Geotechnical Engineering, Structural Engineering, Transportation Engineering, and Water Resources Engineering, or other related fields.

- **Social and Ethical Responsibility:** Committed members of professional and other allied organizations engaging in community development and nation-building.
- **Lifelong Learning:** Involved in continuous training and development in current trends and advancements in their fields of specialization.

The relationship between the SOs and the PEOs is established through outcome mapping. The attainment of the SOs upon graduation provides the knowledge, skills, and professional competencies necessary for graduates to achieve the PEOs within three to five years after graduation. As graduates progress in their professional practice, they build upon these competencies to realize the expected career and professional accomplishments embodied in the PEOs.

The mapping of the SOs to the PEOs is presented in Table 2-1.

Table 2-1. *Mapping of Student Outcomes to Program Educational Objectives*

Student Outcomes, SOs		Program Educational Objectives, PEOs		
		PEO1	PEO2	PEO3
1	Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization to the solution of complex civil engineering problems **(Knowledge Competence)**.	✓		
2	Conduct investigations of complex civil engineering problems using research-based knowledge and research methods, including design of experiments, analysis, and interpretation of data, and synthesis of information to provide valid conclusions **(Investigation).**	✓		
3	Design solutions for complex civil engineering problems and design systems, components or processes that meet desired needs within realistic constraints such as economic, environmental,	✓		

	social, political, ethical, health and safety, manufacturability, cultural, and sustainability, in accordance with standards **(Design/Development of Solutions).**			
4	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings **(Leadership and Teamwork).**	✓	✓	
5	Identify, formulate, research literature, and analyze complex civil engineering problems, reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences **(Problem Analysis).**	✓		
6	Apply ethical principles and commit to professional ethics, and responsibilities and norms of engineering practice **(Ethics and Professionalism).**		✓	✓
7	Communicate effectively on complex civil engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions **(Communication).**		✓	
8	Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex civil engineering problems in a global, economic, societal and environmental context **(Environment and Sustainability).**		✓	
9	Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change **(Lifelong Learning).**			✓

10	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues, and the consequent responsibilities relevant to professional engineering practice and solutions to complex civil engineering problems **(The Engineer and Society).**		✓	
11	Create, select and apply appropriate techniques, skills, resources, and modern engineering and IT tools, including prediction and modelling, to complex civil engineering problems with an understanding of the limitations **(Modern Tool Usage).**	✓		✓
12	Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments **(Project Management and Finance).**		✓	

Graduates are expected to achieve the PEOs after several years of professional experience. Therefore, the SOs must ensure that graduating students are equipped with the knowledge, skills, and competencies necessary to achieve the PEOs as they pursue their professional careers.

2.2 Formulation, Review, and Revision of Student Outcome

2.3.1 Historical Development and Basis of Student Outcomes

The SOs of the BSCE program were **initially formulated in 2018** as part of the program's preparation for outcomes-based accreditation and alignment with national and international standards. This was initially termed Program Outcomes (POs). The formulation was primarily anchored on the provisions of Commission on Higher Education through CHED CMO 92 s. 2017, which defines the expected competencies of engineering graduates in the Philippines.

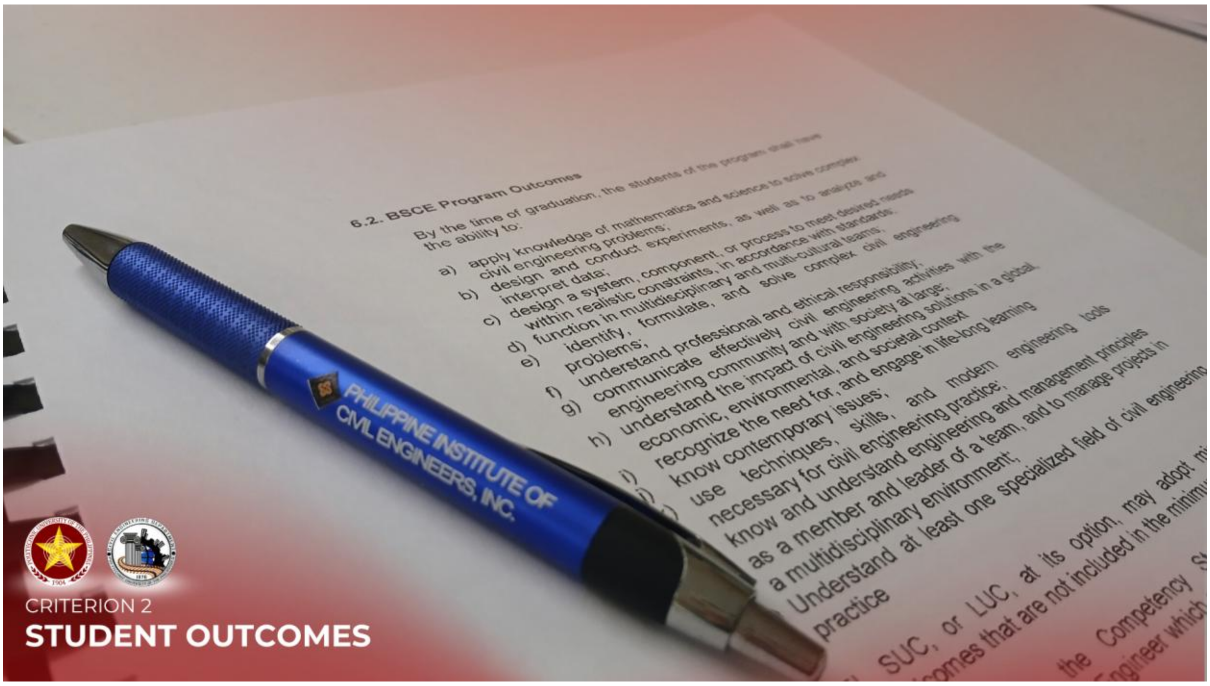

Figure 2-3. *BSCE Program Outcomes as prescribed in CHED Memorandum Order No. 92, s. 2017, Section 6.2. This framework establishes the initial twelve outcomes baseline necessary for compliance with national engineering education mandates and outcomes-based education guidelines. It provided the foundational graduate competencies that guided the program's initial alignment toward national and international accreditation standards.*

The adoption of CHED-prescribed outcomes ensured that the program complied with national educational requirements while establishing a foundation consistent with Outcomes-Based Education (OBE) framework. The resulting twelve (12) SO’s reflect the knowledge, skills, and professional attributes expected of graduates upon completion of the program

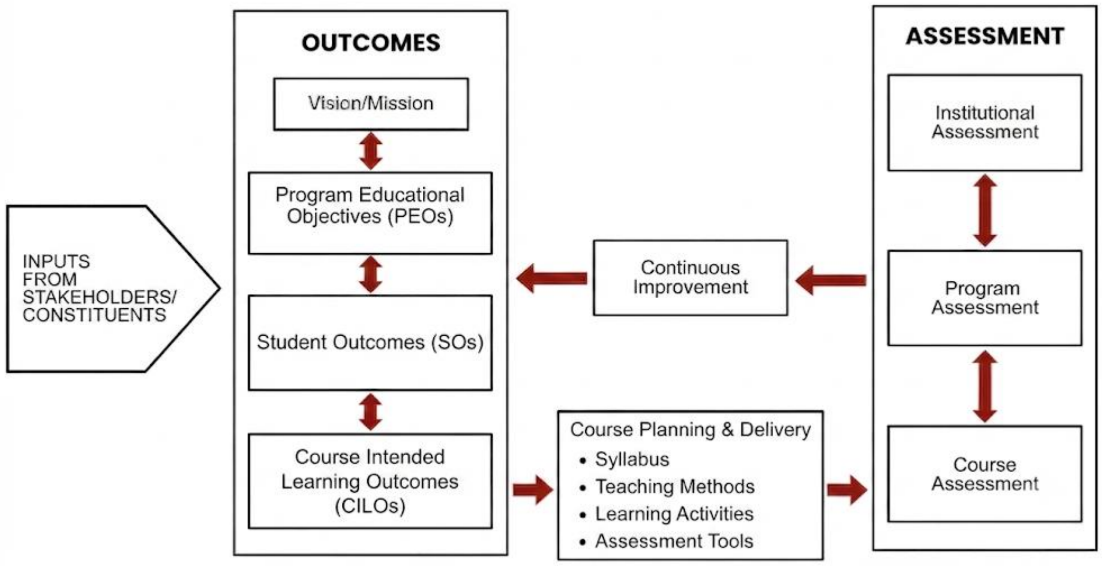

Figure 2-4. *The figure shows the OBTL framework for the BSCE program. It begins with inputs from stakeholders/constituents, which guide the institution's Vision/Mission, PEOs, SOs, and CILOs. These outcomes direct course planning and delivery through the syllabus, teaching methods, learning activities, and assessment tools. Student achievement is measured through course, program, and institutional assessments, while the results are used for continuous improvement to enhance the curriculum, instruction, and overall program quality.*

The development of the Student Outcomes was driven by two primary considerations:

1. **Implementation of Outcomes-Based Education**
   The shift toward OBE required clearly defined, measurable outcomes that focus on student learning rather than content delivery. The SOs serve as the backbone of curriculum design, teaching strategies, and assessment processes.
2. **Preparation for Future Accreditation**
   The program's intent to secure local and international accreditation. Establishing well-defined Student Outcomes (Program Outcomes) ensured readiness for accreditation evaluation and continuous quality improvement.

In the course of continuous quality improvement, the BSCE program undertook a significant academic review that aligned both its curriculum and outcome frameworks. Originally anchored on the 2018 curriculum, the program recognized the need to update its structure to better reflect evolving educational standards and industry demands. This led to the approval of the **Revised 2022 Curriculum for the BSCE program** (Attachment C2-10: Approved Revised 2022 Curriculum for the Bachelor of Science in Civil Engineering (BSCE) Program).

Alongside this curricular revision, a key terminological and conceptual shift was implemented. The term *Program Outcomes* was formally replaced with *Student Outcomes* during the 2022 revision. This change emphasized a more learner-centered perspective, focusing on what students are expected to demonstrate upon completion of the program. The adoption of *Student Outcomes* reflects alignment with global accreditation practices and highlights the program's commitment to producing competent, industry-ready graduates under the Approved Revised 2022 Curriculum.

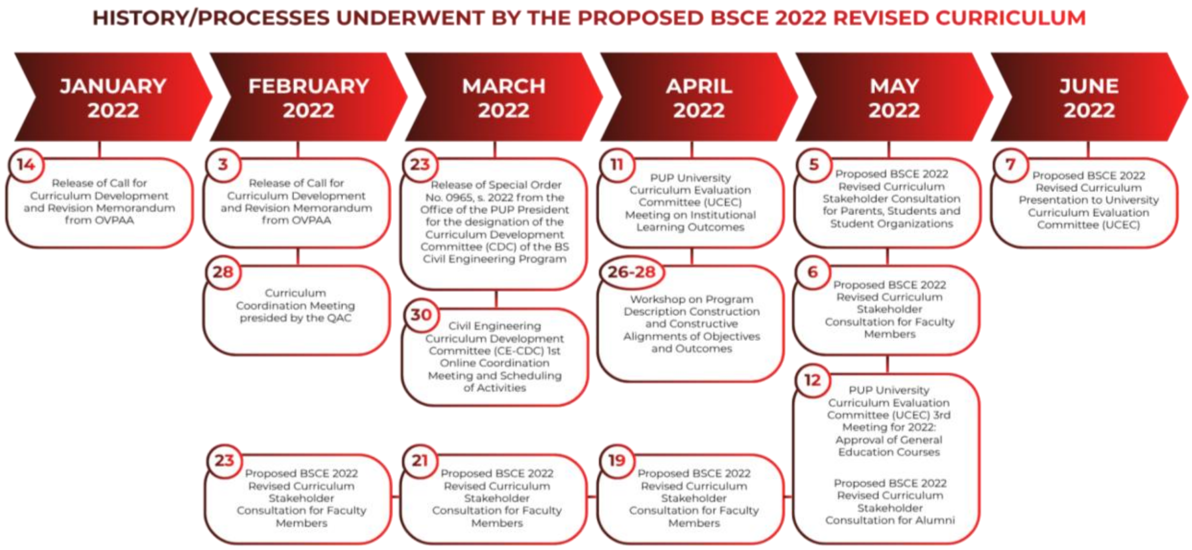

Figure 2-5. *The timeline illustrates the systematic development and approval of the BSCE 2022 Revised Curriculum, highlighting a series of consultations, workshops, and evaluations conducted from January to June 2022. Central to this process was the parallel revision of Program Outcomes, which were redefined and adopted as Student Outcomes to align with contemporary, learner-centered standards. This integration ensured that the updated curriculum is outcomes-based, reflecting clearly defined competencies that students are expected to achieve under the Approved Revised 2022 Curriculum.*

Subsequently, the year 2024 marked a pivotal phase of formal revision, stakeholder consultation, and official institutional approval for the Bachelor of Science in Civil Engineering, Program Educational Objectives.

To meet the dynamic demands of the civil engineering profession and integrate vital feedback regarding required industry competencies, the department initiated a structured review of its program objectives. Prior to the formal launch, a preliminary meeting of GIAB members was held on April 8, 2024, to lay the groundwork for the review process (Attachment C2-05 – GIAB Minutes of Meeting dated April 8, 2024). Afterwards, this process formally began on April 11, 2024, when the Government-Industry Advisory Board (GIAB) convened its first official meeting (Attachment C2-06 – GIAB Minutes of Meeting dated April 11, 2024). Composed of academic administrators, industry experts, specialists across various civil engineering fields, and alumni representatives, as observed in Table 2.3.1.1, the Board evaluated the existing educational objectives against current technological trends and global practices. During this consultation, the GIAB provided critical recommendations that were meticulously incorporated by the University Curriculum Evaluation Committee (UCEC) to refine the objectives, ensuring that future graduates remain globally competitive and responsive to societal needs (Attachment Attachment C2-09 – UCEC Minutes of Meeting dated July 19, 2024; and Attachment C2-04 – Endorsement for the BOR Approval of College of Engineering Programs' PEOs). These continuous efforts extended into 2025 and remain ongoing up to the present (Attachment C2-08 – GIAB Minutes of Meeting dated May 30, 2025; and Attachment C2-07 – GIAB Minutes of Meeting dated March 16, 2026).

Table 2-2. Government Industry Advisory Board for BS Civil Engineering Program in 2024

Name	GIAB Representation	Company	Alumni/ Industry/ Prof. Organization	E-Mail Address
Dr. Emmanuel C. De Guzman	Chair	Polytechnic University of the Philippines	VPAA	ecdeguzman@pup.edu.ph vpaa@pup.edu.ph
Dr. Remedios Ado	Co-chair	Polytechnic University of the Philippines	Dean	rgado@pup.edu.ph
Engr. Kenneth Bryan M. Tana	Member, Secretariat	Polytechnic University of the Philippines	Chairperson, Civil Engineering Department	kbmtana@pup.edu.ph
Engr. Marvin M. Gagarra	Member, PICE Representative	Commission on Audit	Professional Organization	marvin.gagarra2@gmail.com
Engr. Jayr R. Igama	Member, Alumni Representative	BASF (China) Company Ltd.	Alumnus	jay.igama@basf.com

Engr. Divina R. Glino	Member, Alumni Representative, CEM Specialist	Arcadis Consulting Middle East Ltd.	Industry	divine.glino@gmail.com
Engr. Peniel John V. Sta. Maria	Member, Structural Engineering Specialist	J. F. Cancio and Associates	Industry	penielstamaria@gmail.com
Engr. John Jowhell Villegas	Member, Transportation Engineering Specialist	Antipolo Institute of Technology	Industry	johnjowhell.villegas@aitech.edu.ph
Engr. Abner Bondoc	Member, Geotechnical Engineering Specialist	Reinforced Earth Company Philippines	Industry	abner.bondoc@reinforcedearth.com
Dr. Donamel M. Saiyari	Member, Water Resource Engineering Specialist	Department of Science and Technology	Industry	dmsaiyari@pup.edu.ph
Engr. Jonathan V. Aquino	Member, Alumni Representative	Katahira & Engineers Asia Inc	Alumnus	jonathan.v.aquino@gmail.com
Engr. Ramir M. Cruz	Member, Faculty representative	Polytechnic University of the Philippines	Faculty (Former Chairperson, CE Dept.)	rm.cruz@pup.edu.ph
Tristan R. Suayan	Member, Student Representative	Polytechnic University of the Philippines	Student	tristansuayan@gmail.com

Following the integration of stakeholder feedback, the newly revised PEOs and SOs received formal validation and endorsement from the GIAB. The institutional approval process culminated on September 27, 2024, when the highest governing body of the university officially ratified and enacted the current PEOs through Board Referendum No. 162, Series of 2024 (Attachment C2-02 – Board Referendum No. 162 s.2024). This milestone established the three core professional expectations for BSCE alumni three to five years post-graduation, which are seamlessly aligned with the university's updated Vision and Mission (Attachment C2-01 – Board Resolution No. 3587).

2.3.2 Alignment with International and National Frameworks

From its inception, the SOs were deliberately aligned with internationally recognized engineering education frameworks. These include the graduate attributes under the Washington Accord and the SOs prescribed by the Philippine Technological Council through its accreditation arm, PTC ACBET.

This alignment ensured that the program’s graduates possess competencies comparable to those of internationally accredited engineering programs. The mapping of CHED, PTC, and institutional Student Outcomes as shown in Table 2-3 demonstrates consistency and benchmarking across these frameworks.

Table 2-3. *Mapping of PTC, CHED, and PUP student outcomes.*

PHILIPPINE TECHNOLOGICAL COUNCIL Student Outcomes	POLYTECHNIC UNIVERSITY OF THE PHILIPPINES Student Outcomes	COMMISSION ON HIGHER EDUCATION Program Outcomes
STUDENT OUTCOME 1		
Apply knowledge of mathematics, natural science, engineering fundamentals, and an	**Apply knowledge of mathematics, natural science, engineering fundamentals and an**	*Ability to apply knowledge of mathematics and sciences to solve Industrial Engineering problems.*

engineering specialization to the solution of complex engineering problems.	**engineering specialization to the solution of complex civil engineering problems.**	
STUDENT OUTCOME 2		
Conduct investigations of complex engineering problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.	**Conduct investigations of complex civil engineering problems using research-based knowledge and research methods including design of experiments, analysis, and interpretation of data, and synthesis of information to provide valid conclusions.**	*Ability to design and conduct experiments, as well as to analyze and interpret data.*
STUDENT OUTCOME 3		
Design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations	**Design solutions for complex civil engineering problems and design systems, components or processes that meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, cultural, and sustainability, in accordance with standards.**	*Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability, in accordance with standards*
STUDENT OUTCOME 4		
Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.	**Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.**	*Ability to function on multi-disciplinary teams.*

STUDENT OUTCOME 5		
Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.	**Identify, formulate, research literature, and analyze complex civil engineering problems, reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.**	*Ability to identify, formulate, and solve Industrial Engineering problems.*
STUDENT OUTCOME 6		
Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.	**Apply ethical principles and commit to professional ethics, and responsibilities and norms of engineering practice.**	*Understanding of professional and ethical responsibility.*
STUDENT OUTCOME 7		
Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.	**Communicate effectively on complex civil engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.**	*Ability to communicate effectively.*
STUDENT OUTCOME 8		

Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental context.	**Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex civil engineering problems in a global, economic, societal and environmental context.**	*Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context*
STUDENT OUTCOME 9		
Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.	**Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.**	*Recognition of the need for, and an ability to engage in life-long learning.*
STUDENT OUTCOME 10		
Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems.	**Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues, and the consequent responsibilities relevant to professional engineering practice and solutions to complex civil engineering problems.**	*Knowledge of contemporary issues.*
STUDENT OUTCOME 11		
Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering problems with an understanding of the limitations.	**Create, select and apply appropriate techniques, skills, resources, and modern engineering and IT tools, including prediction and modelling, to complex civil engineering problems with an understanding of the limitations.**	*Ability to use the techniques, skills,and modern engineering tools necessary for Civil / Electronics / Industrial Engineering practice.*
STUDENT OUTCOME 12		

Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.	**Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.**	*Knowledge and understanding of engineering management principles as a member and leader in a team, to manage projects and in multidisciplinary environments*

2.3.3 Role of Faculty and Curriculum Development Committee

The formulation process was collaborative, and faculty driven. The department faculty members, under the leadership of the *Department Chairperson*, **Engr. Kenneth Bryan M. Tana** played a central role in contextualizing the CHED-prescribed outcomes to the specific needs of the Civil Engineering program.

Supporting this process was the **CDC Committee**, which provided technical guidance on outcomes formulation, alignment, and assessment. The committee ensured that:

- Outcomes were measurable and aligned with performance indicators
- Mapping with PEOs was coherent
- Assessment mechanisms were consistent with OBE principles

Furthermore, recommendations for review, and revision are initiated through CDC Committee reports and deliberations, ensuring a structured and evidence-based approach to outcomes management.

The revision process is initiated through several *key triggers* that ensure the curriculum remains relevant, compliant, and continuously improving. A scheduled curriculum review provides a structured opportunity to evaluate and update course content, while revisions of SO are aligned with any broader changes in the curriculum to maintain coherence and consistency. Additionally, updates in CHED policies and PTC ACBET requirements necessitate timely adjustments to ensure ongoing compliance with national and accreditation standards. Finally, insights derived from assessment results and continuous improvement recommendations play a crucial role, guiding evidence-based enhancements that strengthen the overall quality and effectiveness of the program.

The revision process actively involves key stakeholders to ensure that all perspectives are considered and that decisions are both strategic and well-informed. The Department Chairperson leads and oversees the process, guiding discussions and ensuring alignment with institutional goals. Faculty members contribute their subject expertise and classroom insights, helping shape meaningful and practical improvements to the curriculum. The CDC plays a critical role in ensuring that revisions remain aligned with outcomes-based education principles and accreditation standards. Furthermore, the College administration provides oversight and formal approval, ensuring that all proposed changes meet institutional policies and strategic directions.

Key Improvements

- Enhanced alignment with updated CHED and PTC ACBET outcomes
- Improved clarity and articulation of outcome statements
- Strengthened completeness and mapping with PEOs
- Refinement of performance indicators and assessment methods
- The revision process ensured that the Student Outcomes remained current, relevant, and responsive to both academic and industry needs

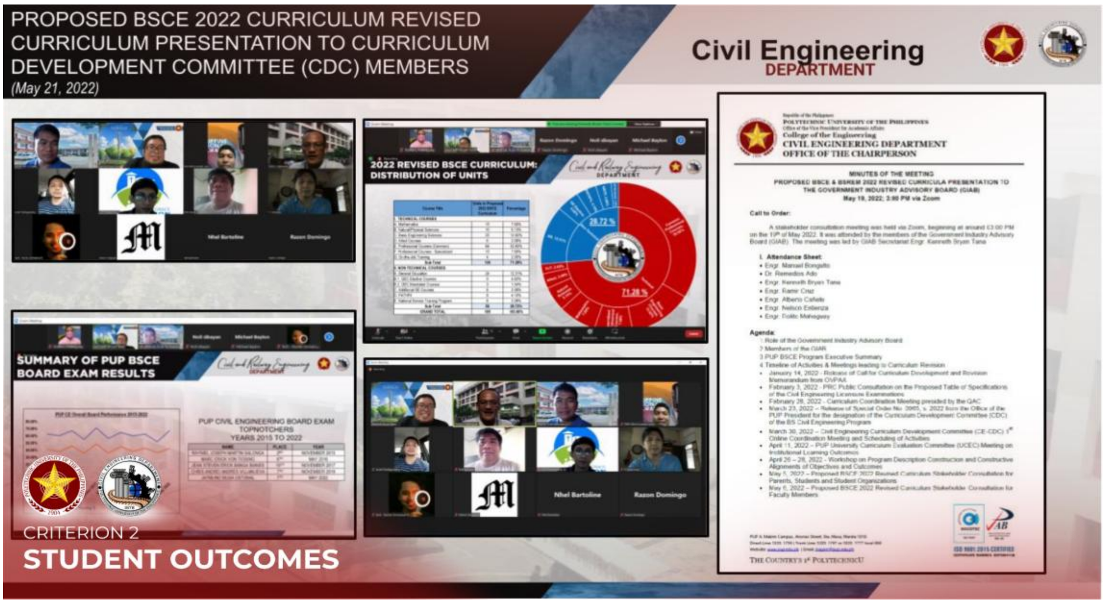

Figure 2-6. *The First Online PUP Civil Engineering Curriculum Development Committee Coordination Meeting was held on March 30, 2022. The Coordination meeting was attended by the CDC Committee members as called by the Memorandum Order CE.002 S. 2022 and Memorandum No. RE 002 s. 2022.*

The revision process actively involves key stakeholders to ensure that all perspectives are considered and that decisions are both strategic and well-informed:

- Faculty Members
- Academe Partners through part-time faculty members who also teach in other National Universities and Colleges

- Industry Partners through GIAB
- Alumni
- Students

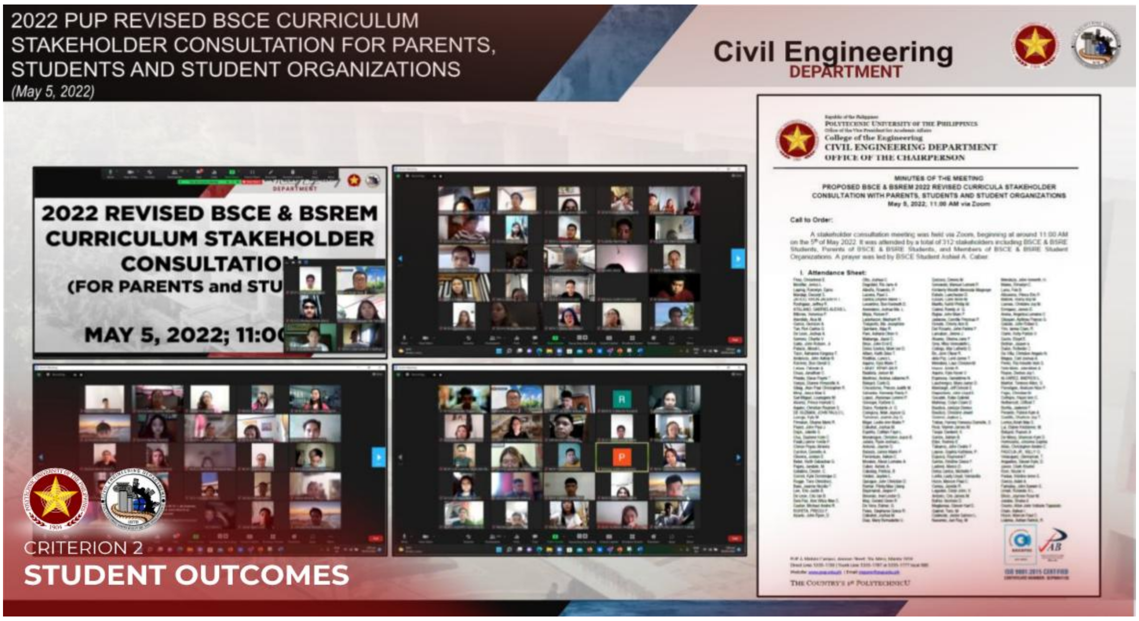

Figure 2-7. *The stakeholder consultation with parents, students, and student organizations for the 2022 Revised BSCE Curriculum highlights the importance of inclusive participation in curriculum development. Through this engagement, valuable perspectives on student needs, learning experiences, and support systems were gathered to refine the Student Outcomes (formerly Program Outcomes).*

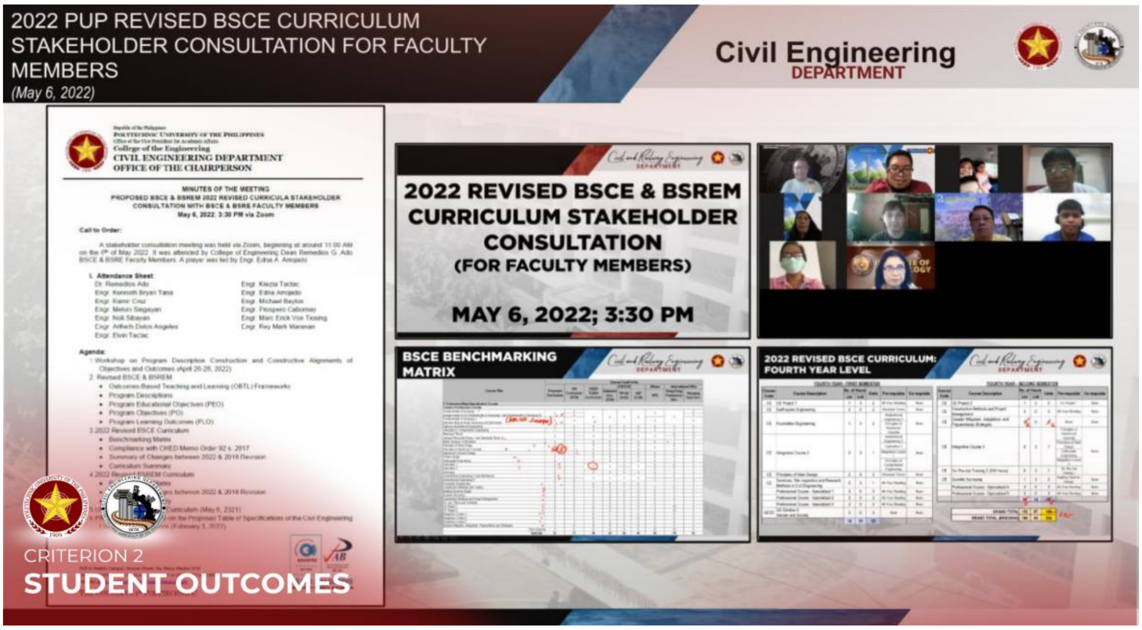

Figure 2-8. *The stakeholder consultation with faculty members for the 2022 Revised BSCE Curriculum emphasizes the critical role of academic expertise in curriculum enhancement. Through collaborative discussions on benchmarking, course alignment, and instructional strategies, faculty members contributed to the refinement of Student Outcomes (formerly Program Outcomes). This ensured that*

the curriculum remains academically rigorous, coherent, and aligned with desired graduate competencies.

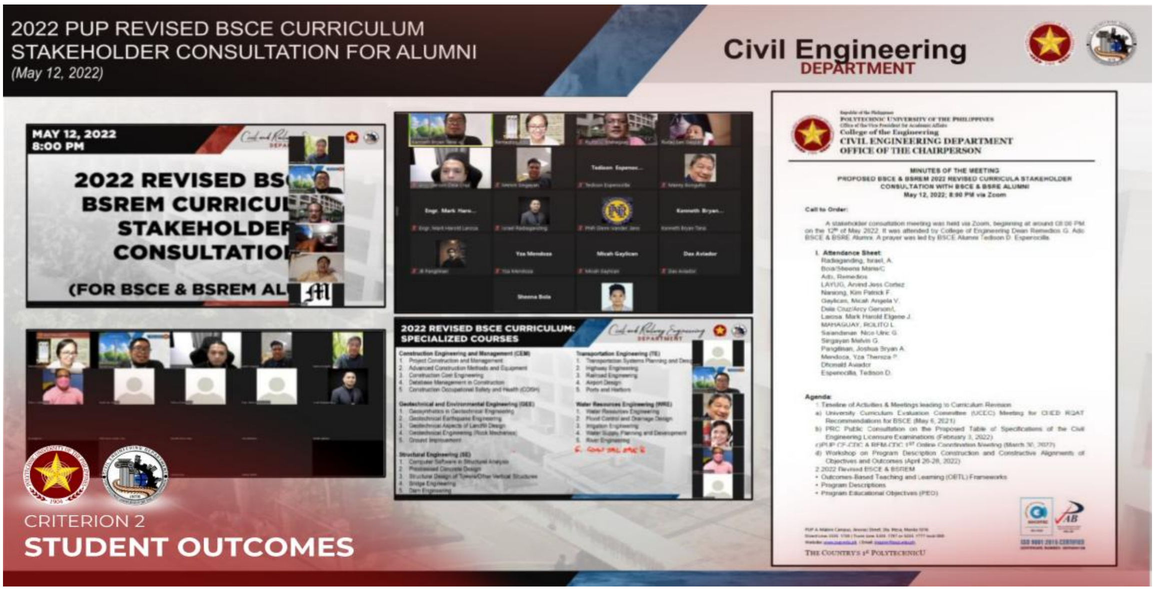

Figure 2-9. *The stakeholder consultation with alumni for the 2022 Revised BSCE Curriculum highlights the importance of graduate insights in enhancing program relevance and effectiveness. Drawing from their professional experiences, alumni contributed valuable feedback that informed the refinement of Student Outcomes (formerly Program Outcomes), ensuring that the competencies outlined in the curriculum remain practical, applicable, and aligned with real-world engineering demands. This engagement reinforces the Approved Revised 2022 Curriculum's commitment to continuous improvement and outcomes-based education grounded in industry and career realities.*

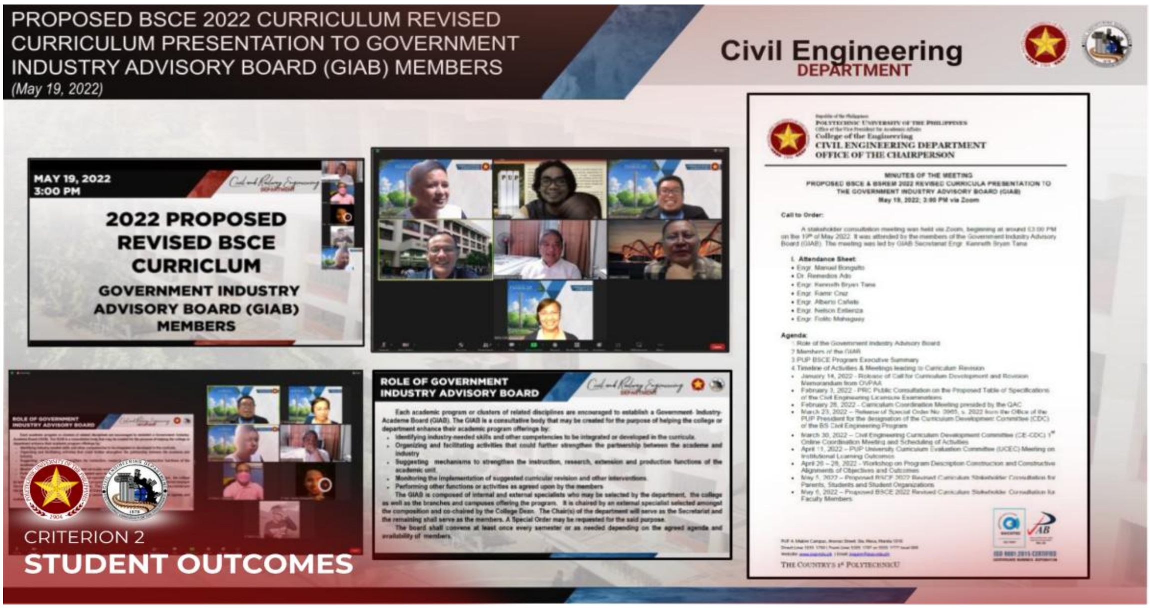

Figure 2-10. *The consultation with the Government Industry Advisory Board (GIAB) for the Proposed BSCE 2022 Revised Curriculum underscores the vital role of industry stakeholders in shaping an outcomes-based education. Through this engagement, feedback from professionals was integrated to refine the Student Outcomes (formerly Program Outcomes), ensuring that the competencies defined in the curriculum are relevant, responsive, and aligned with current industry standards. This collaborative process strengthens the Approved Revised 2022 Curriculum by anchoring student learning on real-world expectations and professional practice.*

2.3.4 Current Status and Continuous Improvement

At present, the SOs are fully institutionalized and are regularly assessed through defined performance indicators and rubrics, reviewed alongside curriculum which updates as necessary based on changes in external standards and internal evaluation

The program maintains a continuous improvement cycle, where feedback from stakeholders and assessment results inform future refinements. This ensures that the SOs remain aligned with global engineering education standards and industry expectations.

2.3.5 Basis and References for Student Outcomes

The SOs of the BSCE program are grounded on multiple authoritative sources to ensure compliance, relevance, and global comparability:

- **Institutional Alignment**: Alignment with the mission and vision of the Polytechnic University of the Philippines ensures that the SOs support:
  - National development goals
  - Professional competence
  - Lifelong learning and ethical responsibility
- **Centers of Excellence (COEs) and Centers of Development (CODs)**: The SOs were reviewed and benchmarked against CHED-recognized COEs and CODs offering Civil Engineering and related engineering programs to ensure alignment with best practices in engineering education.
- **National Regulatory Framework** (**CHED CMO 92 s. 2017)**: Provides the foundational competencies required of Civil Engineering graduates in the Philippines.
- **National Accreditation Body** (**Philippine Technological Council – PTC ACBET)**: Defines SOs aligned with international accreditation systems and ensures quality assurance in engineering education.
- **International Benchmark** (**Washington Accord):** Establishes globally recognized graduate attributes, ensuring that program outcomes meet international standards for engineering practice.

- **Sustainable Development Goals (SDGs):** The SOs promote sustainable, ethical, and socially responsible engineering practices that contribute to the achievement of the United Nations Sustainable Development Goals.

2.3.6 Benchmarking and External Alignment

The program adopts a benchmarking and alignment process to ensure that its SOs are consistent with national and international standards. The SOs were developed with reference to CHED policies and standards, PTC-ACBET accreditation requirements, recognized international engineering education frameworks, and best practices from CHED-recognized COEs and CODs.

Through this process, the SOs reflect a balanced integration of regulatory requirements, accreditation standards, industry expectations, SDGs, and institutional objectives. This alignment ensures that graduates are prepared to address complex engineering challenges and contribute effectively to both national development and global engineering practice.

2.4. Process for Formulation, Review, and Revision

The institution adopts a **systematic and cyclical process** for the formulation, review, and revision of curriculum alongside with SOs (Attachment C2-12 – Curriculum Development/Revision Guidelines in 2026). This process ensures alignment with CHED, PTC, and WA standards, participation of relevant stakeholders, data-driven decision-making through assessment results and continuous enhancement of program quality.

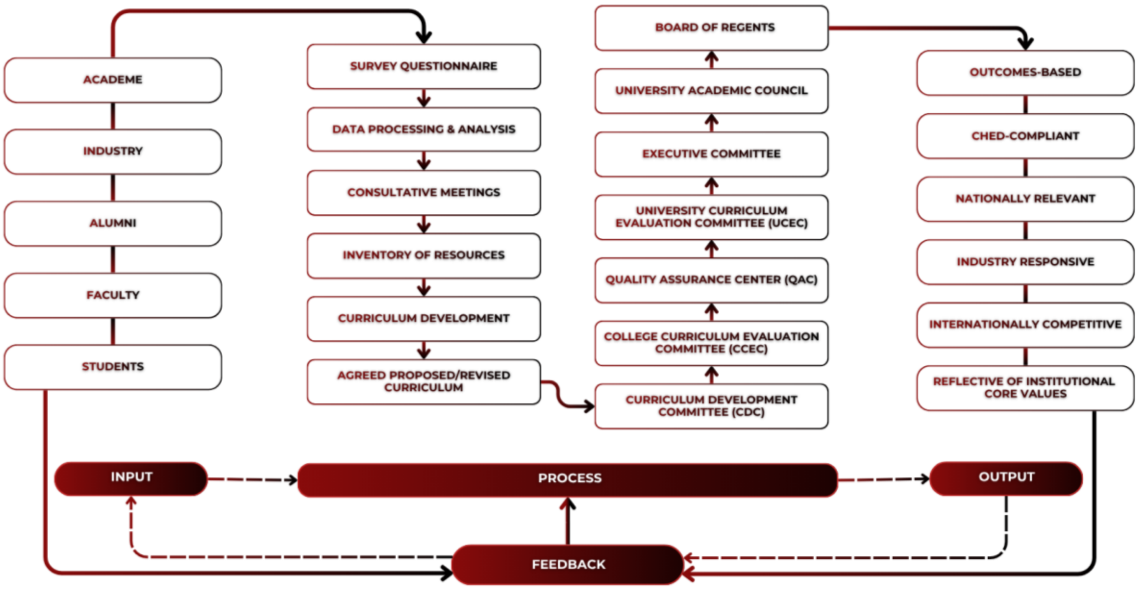

Figure 2-11. Curriculum Development/Revision Process Framework in 2026. *The framework illustrates the University's curriculum development and review process, which integrates benchmarking, stakeholder consultation, curriculum evaluation, quality assurance review, and multi-level approval mechanisms. It ensures that curriculum revisions are systematically developed, assessed, and aligned with academic, professional, and industry requirements.*

Ultimately, this structured mechanism **demonstrates a strong commitment to Continuous Quality Improvement (CQI)** and ensures that graduates are equipped with competencies that meet both national and global expectations.

2.4.1 Formulation Process of Student Outcomes

The SOs are formulated to define the competencies, knowledge, skills, and attitudes that graduates are expected to demonstrate upon completion of the program. These outcomes ensure alignment with national and international standards and serve as the foundation for OBE implementation.

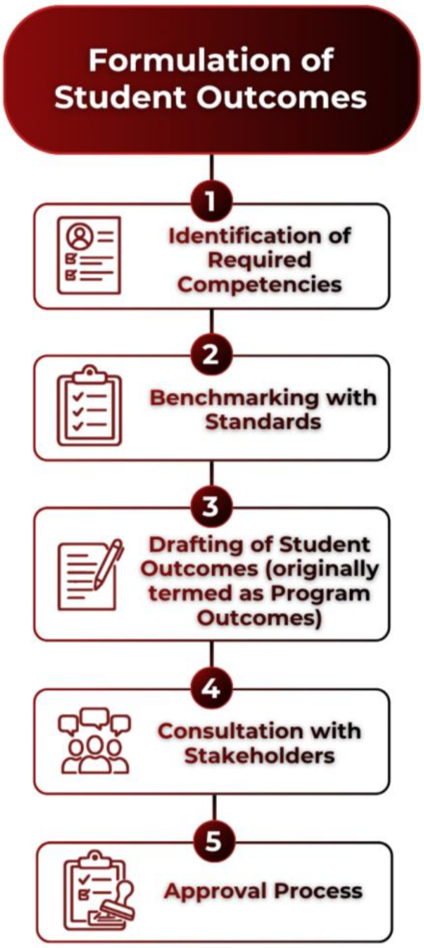

Figure 2-12. *This flowchart illustrates the systematic, evidence-based approach used to establish graduate competencies through industry benchmarking and stakeholder consultation. The final stage outlines the formal multi-level approval pipeline required to ensure institutional alignment and rigorous quality assurance.*

The formulation of SOs follows a systematic and evidence-based process:

- **Identification of Required Competencies**: The process begins with identifying the essential competencies expected from graduates based on program objectives, industry needs, and professional practice requirements.
- **Benchmarking with Standards**: The identified competencies are reviewed and benchmarked against CHED policies and standards, PTC-ACBET Student Outcomes, recognized international engineering education frameworks, and best practices from CHED-recognized COEs and CODs.
- **Drafting of Student Outcomes** (originally termed as Program Outcomes): The faculty members, in collaboration with the CDC, draft the SOs. The drafting process ensures clarity, measurability, and alignment with program educational objectives.
- **Consultation with Stakeholders**: Where applicable, consultations are conducted with relevant stakeholders such as industry partners, alumni, and advisory boards to validate the relevance and applicability of the SOs.
- **Approval Process**: The finalized SOs undergo a formal approval process to ensure institutional alignment and quality assurance:
    - Curriculum Development Committee (CDC)
    - College Evaluation Curriculum Committee (CECC)
    - Quality Assurance Center (QAC)
    - University Evaluation Curriculum Committee (UECC)
    - Executive Committee (ExeCom)
    - University Academic Council (UAC)
    - Board of Regents (BOR)

2.4.2 Review Mechanism of Student Outcomes

The SOs are regularly reviewed to ensure their continued relevance, responsiveness, and alignment with institutional goals, industry needs, professional standards, and national and international quality assurance requirements. This review process supports the University's commitment to OBE and CQI.

The review of SOs is conducted whenever significant changes arise from CHED policies and standards, accreditation requirements, industry trends, stakeholder feedback, curriculum reviews, tracer studies, and other program assessment activities.

The review process is led by the CDC in consultation with faculty members, industry representatives, alumni, students, and other relevant stakeholders. Recommendations are evaluated through the University's curriculum review and approval process to ensure academic quality and institutional alignment.

Assessment data collected from course outcomes, performance indicators, and evaluation tools are analyzed to determine the extent of SO attainment. These results are used to identify strengths and gaps in student performance, provide evidence-based recommendations and support decision-making for improvements.

This structured use of assessment results ensures that identified gaps are addressed, thereby closing the loop and directly supporting CQI.

2.4.3 Revision Process of Student Outcomes

The revision process highlights the institution’s commitment to continuous improvement by ensuring that Student Outcomes are regularly enhanced based on assessment results, stakeholder feedback, and evolving standards. This is a critical component for accreditation as it demonstrates an active and functioning CQI system.

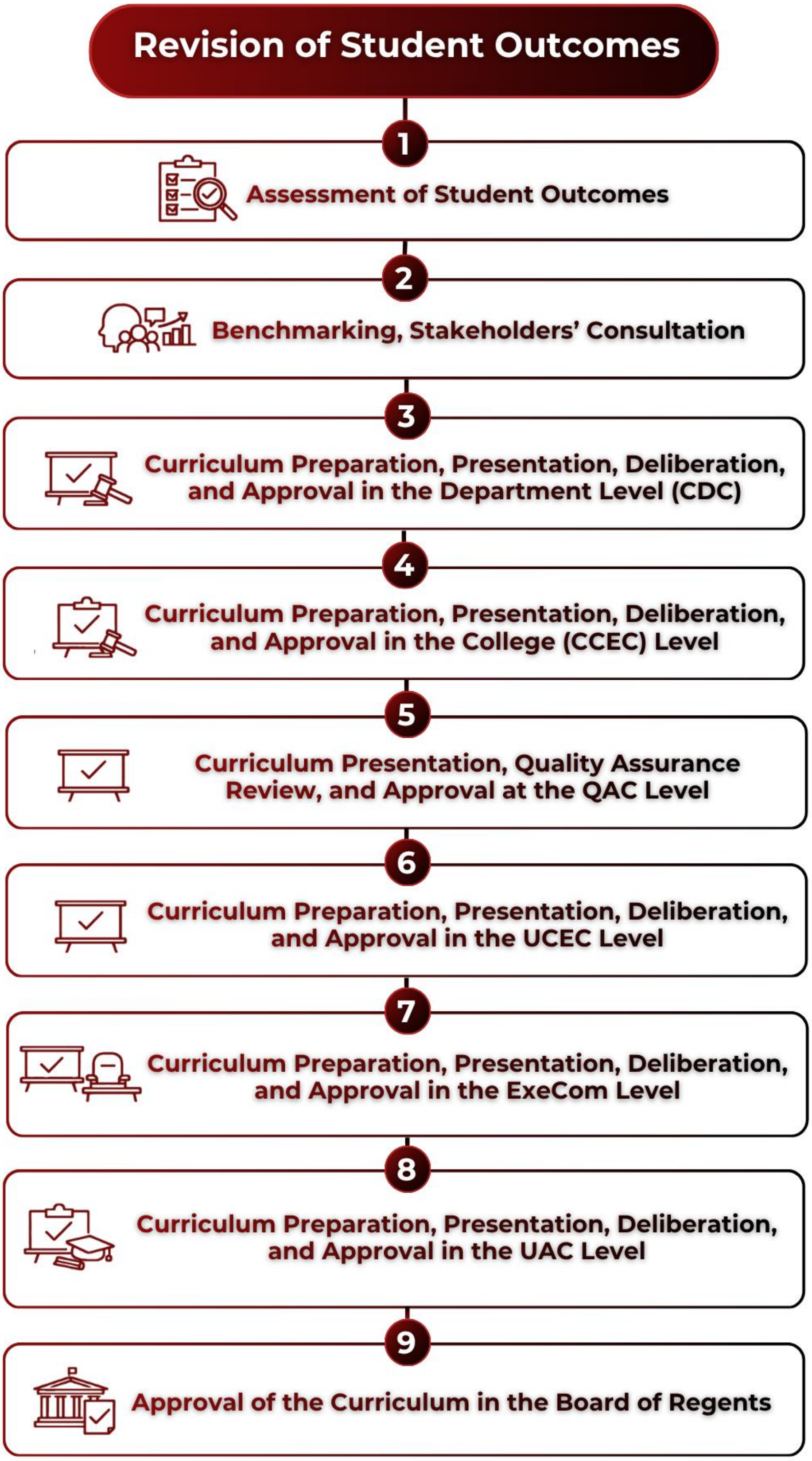

Figure 2-13. *This diagram outlines the systematic procedure for updating and refining student outcomes based on annual performance indicators and consolidated CDC feedback reports. The process highlights the implementation of curricular actions and the formal administrative approval required to officially adopt revised outcomes.*

The following are the processes involved in the revision of Student Outcomes:

- **Annual Assessment of Student Outcomes**: SOs are assessed annually using:
    - Defined performance indicators
    - Standardized rubrics
    - Systematic analysis of assessment results
- **Feedback Loop and Reporting**: The CDC consolidates and analyzes assessment findings and prepares an annual report. This report includes:
    - Level of attainment of SOs
    - Identified gaps and areas for improvement
    - Recommended actions
- **Actions Taken**: Based on the CDC report, the following actions may be implemented:
    - Revision or refinement of Student Outcomes
    - Curriculum enhancement or modification
    - Improvement of teaching-learning strategies
- **Internal Deliberation and Approval**: Proposed revisions undergo internal deliberation and are subjected to the institutional approval process:
    - **Curriculum Development Committee**: Develops and revises the curriculum, conducts benchmarking and stakeholder consultations, and prepares the curriculum proposal and supporting documents.
    - **College Evaluation Curriculum Committee**: Reviews, evaluates, and enhances the curriculum proposal at the college level before endorsement for quality assurance review.
    - **Quality Assurance Center**: Verifies the completeness, compliance, and quality of curriculum proposals and endorses compliant proposals for higher-level evaluation.
    - **University Evaluation Curriculum Committee**: Conducts university-level review and deliberation of curriculum proposals to ensure alignment with institutional, academic, and quality standards.
    - **Executive Committee**: Reviews and approves curriculum proposals from an executive and strategic management perspective before endorsement to the Academic Council.
    - **University Academic Council**: Evaluates and approves academic aspects of the proposed curriculum to ensure consistency with the University's academic policies and standards.
    - **Board of Regents**: Serves as the highest approving authority for new and revised curricula and authorizes their implementation.

- **Documentation of Improvements**: All revisions and actions taken are properly documented to provide evidence of continuous improvement and compliance with accreditation requirements.

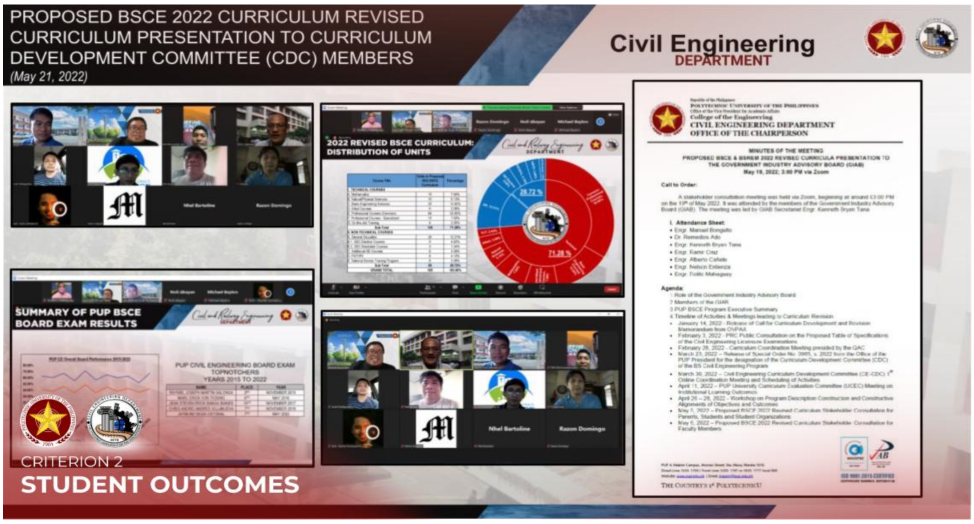

Figure 2-14. *The presentation of the Proposed BSCE 2022 Revised Curriculum to the CDC highlights the collaborative review of curriculum mapping, program rationale, and alignment strategies. Integral to this process is the refinement of Program Outcomes, formally transitioned into Student Outcomes, ensuring that all course components and learning experiences are aligned with clearly defined, measurable competencies.*

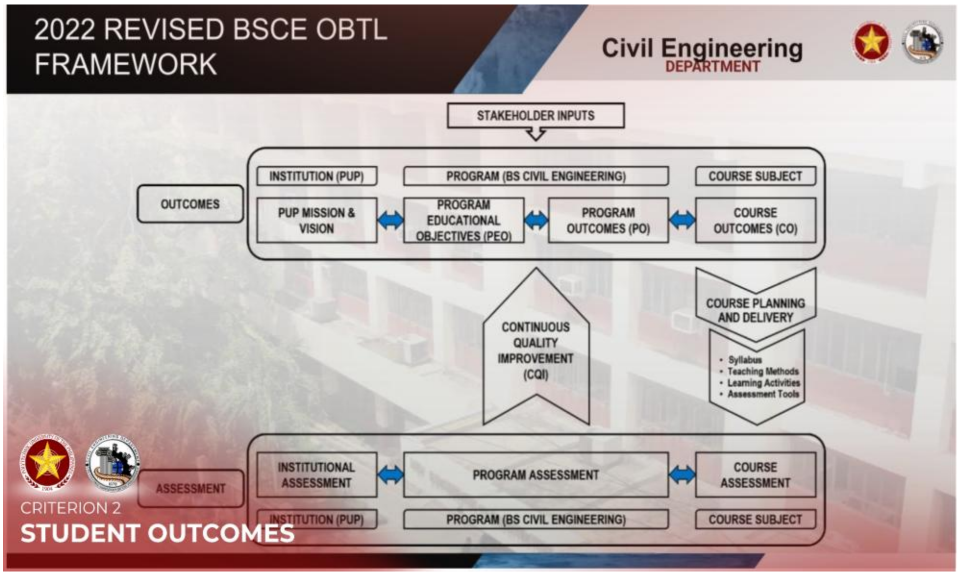

Figure 2-15. *The 2022 Revised BSCE OBTL Framework outlines the system for outcomes-based education and quality assurance. The top section shows how the institutional mission connects directly to PEOs, POs, and Course Outcomes (COs). The bottom section aligns these outcomes with institutional, program, and course assessments, all driven by a CQI process to guide curriculum delivery*

2.4.4 Continuous Quality Improvement and Implementation

The program adopts a systematic and evidence-based approach to **Continuous Quality Improvement**, ensuring that assessment results are effectively utilized to enhance student learning outcomes and program quality. This process is anchored on an annual cycle as shown in the following figure, which enables the program to remain responsive to academic and industry needs:

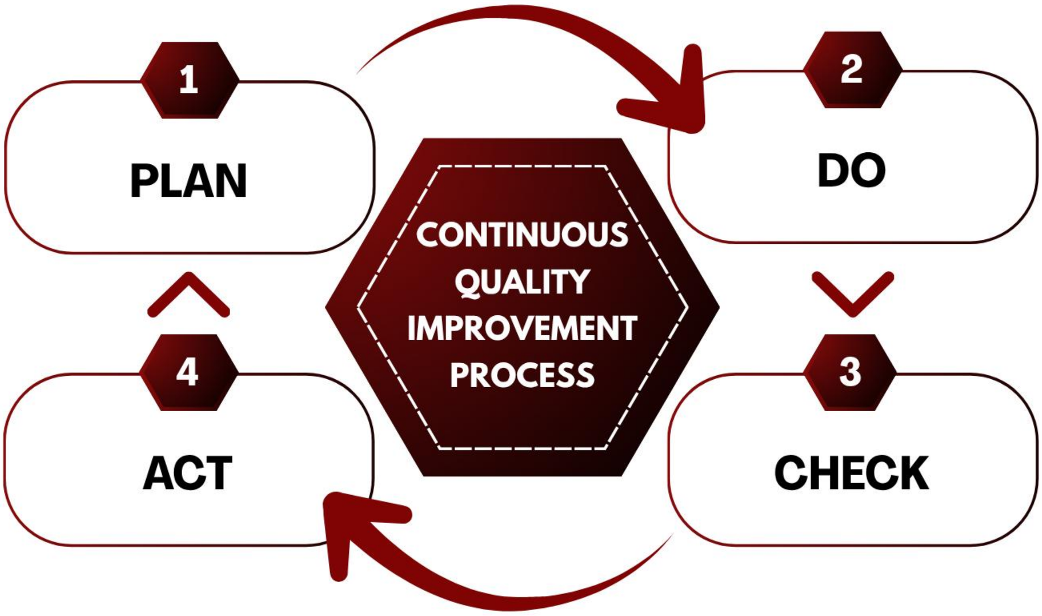

Figure 2-16. *The figure outlines an annual, four-stage CQI cycle designed to systematically enhance student learning outcomes and program quality. The process begins with faculty collecting objective performance data for analysis by the Curriculum Development Committee, which then implements and monitors targeted actions—such as updated course content and teaching strategies—to complete the quality loop.*

The BS Civil Engineering program adopts the **Plan–Do–Check–Act (PDCA) Cycle** as its framework for Continuous Quality Improvement to ensure the sustained attainment and relevance of the Student Outcomes (Attachment C2-13 – 2018-2022 CQI for Student Outcomes Assessment). The following are the PDCA Cycle processes:

- **Plan.** Student Outcomes, performance indicators, assessment tools, and evaluation criteria are established and aligned with program objectives, accreditation requirements, stakeholder expectations, and institutional goals.

- **Do.** Annual assessment is conducted using performance indicators and standardized rubrics aligned with the Student Outcomes. Faculty members collect, evaluate, and consolidate assessment data from designated courses and learning activities at the end of each academic term.

- **Check.** The assessment results are systematically analyzed to determine the level of Student Outcome attainment. Trends, strengths, and areas for improvement are identified through the review of assessment data, stakeholder feedback, and other relevant evidence. The Curriculum Development Committee facilitates the evaluation process and validates the findings.

- **Act.** Based on the results of the evaluation, appropriate improvement actions are implemented. These may include curriculum enhancement, revision of course content, improvement of teaching and assessment strategies, faculty development activities, and strengthening of learning resources. The effectiveness of these actions is monitored and evaluated in succeeding assessment cycles to ensure continuous improvement and sustained quality of the program.

The CQI process, guided by the PDCA Cycle, enables the BS Civil Engineering program to systematically assess, evaluate, and enhance the attainment of its Student Outcomes. Through regular assessment, data-driven evaluation, and the implementation of appropriate improvement actions, the program ensures that its curriculum and educational practices remain responsive to stakeholder needs and accreditation requirements. This continuous cycle of planning, implementation, evaluation, and improvement supports the sustained quality and relevance of the program.

2.4.5 Documentation and Evidence

To support the effective implementation of Outcomes-Based Education and Continuous Quality Improvement processes, the program maintains comprehensive and well-organized documentation and evidence. These records serve as verifiable proof that assessment, evaluation, and improvement activities are systematically conducted, properly reviewed, and consistently implemented.

A key component of documentation includes **minutes of meeting**, which capture discussions, decisions, and action plans from faculty meetings and sessions led by the CDC. These minutes provide a clear record of how assessment results are analyzed and how improvement strategies are agreed upon and monitored over time.

All approved revisions to SOs are formally documented, including the rationale for changes, supporting data, and evidence of stakeholder consultation when applicable. This ensures transparency and traceability in how the program adapts its outcomes to align with evolving academic standards and industry requirements.

In addition, curriculum review records are systematically archived. These include documentation of curriculum mapping, course revisions, alignment of courses with SOs, and validation of changes through appropriate academic bodies**.** Such records demonstrate that curriculum enhancements are directly linked to assessment findings and CQI actions.

The program further ensures that relevant information is disseminated through official publications, such as the institutional website, course syllabi, student handbooks, and other academic materials. These platforms reflect updated Student Outcomes, course content, and assessment methods, ensuring that stakeholders are informed of current program expectations and standards.

Collectively, these documents establish a transparent and auditable system that evidences the full CQI cycle and confirms that continuous improvement efforts are not only implemented but also properly recorded, reviewed, and communicated.

2.5 Performance Indicators for each of the Student Outcomes

The BSCE program of the Polytechnic University of the Philippines has accepted and implemented the term **Performance Indicators (PI)** as described in the CHED CMO 92 s. 2017. Performance Indicators are specific and measurable statements identifying the performance required to meet the outcomes, confirmable through evidence

Table 2-4. *Performance Indicators for each of the Student Outcomes*

Student Outcomes	Performance Indicator
Student Outcome 1	Application of knowledge of mathematical models or equations to solve engineering problems.

Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization to the solution of complex civil engineering problems **(Knowledge Competence)**.	Application of knowledge of natural and engineering science to solve complex civil engineering problems.
	Application of knowledge of engineering fundamentals to the solution of complex civil engineering problems.
Student Outcome 2 Conduct investigations of complex civil engineering problems using research-based knowledge and research methods including design of experiments, analysis, and interpretation of data, and synthesis of information to provide valid conclusions **(Investigation)**.	Conduct laboratory experiments or fieldwork in accordance with manuals and engineering standards while maintaining orderliness;
	Design the experimental plan of a laboratory experiment or fieldwork;
	Analyze the collected data from laboratory experiment and field works using appropriate methods in accordance with engineering principles
	Interpret the collected data from laboratory experiment and field works and draw out conclusions using appropriate methods in accordance to engineering principles
Student Outcome 3 Design solutions for complex civil engineering problems and design systems, components or processes that meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, cultural, and sustainability, in accordance with standards **(Design/Development of Solutions)**.	Formulate criteria evaluation and analyze realistic constraints of a design problem
	Create a design project/ prototype/ system/ component/ process that meets the relevant design codes and standards
Student Outcome 4 Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings **(Leadership and Teamwork)**.	Participate, express ideas freely and execute member tasks towards attainment of the objectives. (idea generation)
	Integrate input from team members and make decisions based on objective criteria. (decision making)
Student Outcome 5 Identify, formulate, research literature, and analyze complex civil engineering problems, reaching	Identify complex civil engineering problems and identify the key issues, factors, and concepts affecting the problem;

substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences **(Problem Analysis)**.	Analyze and formulate solutions to complex civil engineering problems;
Student Outcome 6 Apply ethical principles and commit to professional ethics, and responsibilities and norms of engineering practice **(Ethics and Professionalism)**.	Demonstrate ethical and professional behavior during practice
	Demonstrate professionalism by complying to relevant codes and standards
Student Outcome 7 Communicate effectively on complex civil engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions **(Communication)**.	Use effective written communication by writing grammatically correct and logical reports using appropriate graphics and technical style.
	Use effective oral communication by presenting and defending ideas with sound reasoning and in an articulate manner.
Student Outcome 8 Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex civil engineering problems in a global, economic, societal and environmental context **(Environment and Sustainability)**.	Explain the impacts of civil engineering solutions in a global, economic, environmental, and societal context.
	Establish the significance of the solutions in terms of beneficiaries.
Student Outcome 9 Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change **(Lifelong Learning)**.	The student can demonstrate self-directed learning through acquired new knowledge from various modes and strategies (research, attendance in conferences and exhibits, site visits)
Student Outcome 10 Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues, and the consequent responsibilities relevant to professional engineering practice and solutions to complex civil engineering problems **(The Engineer and Society)**.	Identify contemporary issues in a chosen civil engineering field of specialization.
	Determine effect of contemporary issues in the civil engineering discipline.
Student Outcome 11 Create, select and apply appropriate techniques, skills, resources, and modern engineering and IT tools, including prediction and modelling, to complex civil engineering problems with an understanding of the limitations **(Modern Tool Usage)**.	Creation of most appropriate techniques or modern tools for a specific civil engineering task.
	Selection of skills relevant to the civil engineering practice using modern engineering tools and techniques.

	Application of the techniques, resources, skills, and modern engineering tools to address complex civil engineering problems.
Student Outcome 12 Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments **(Project Management and Finance)**.	Use appropriate tools and theories to manage multidisciplinary projects in a team.
	Identify and devise solutions to real world problems as a member/leader of a cross-functional team.

2.6 Deployment Process of Student Outcomes

The **Student Outcomes** of the BS Civil Engineering program are made accessible through multiple platforms to ensure transparency and awareness among all stakeholders. These outcomes are publicly available on the **Polytechnic University of the Philippines (PUP)** official website (https://www.pup.edu.ph/CE/BSCE), and are also included in the **program brochure**, **OBTLP**, and **course syllabi**.

To enhance deployment and stakeholder engagement, the following processes are systematically carried out:

- **Advisory Board Meetings** – Student Outcomes are presented and discussed during advisory board meetings attended by key program stakeholders and constituencies to gather feedback and ensure alignment with industry needs.
- **Department-Wide Student Orientation** – At the start of every semester, the department conducts a student orientation for all year levels. This orientation covers institutional and program policies, academic requirements, and includes a detailed presentation of the Student Outcomes to promote awareness and understanding among students.
- **Departmental Faculty Meeting** – Prior to the beginning of each semester, the Department Chairperson convenes a meeting with all faculty members to reiterate the significance of the Student Outcomes. This meeting serves to align teaching strategies and assessment practices with the expected program outcomes.
- **Standardized PowerPoint Template for Faculty** – The department provides a standardized PowerPoint presentation template to be used by all faculty members during the first meeting of each class. This template includes a dedicated section for

the discussion of Student Outcomes to ensure consistent delivery across all courses (Attachment C2-11 – PowerPoint Presentation of Class Orientation).

- **Classroom-Level Discussions** – Faculty members emphasize Student Outcomes on the first day of every class, helping students understand how course objectives and activities contribute to achieving these outcomes.
- **Bulletin Board Displays** – Student Outcomes are posted on designated bulletin boards in the College of Engineering for continuous visibility and reference by students and visitors.
- **OBTLP Syllabus** – The Student Outcomes are presented in all OBTLP syllabus in all subjects (Attachment C2-03 – Sample Syllabus).

Figure 2-17. *BSCE Student Outcomes are displayed on designated bulletin boards of Civil Engineering Laboratory in the College of Engineering to maintain consistent visibility and serve as a reference for students and visitors.*

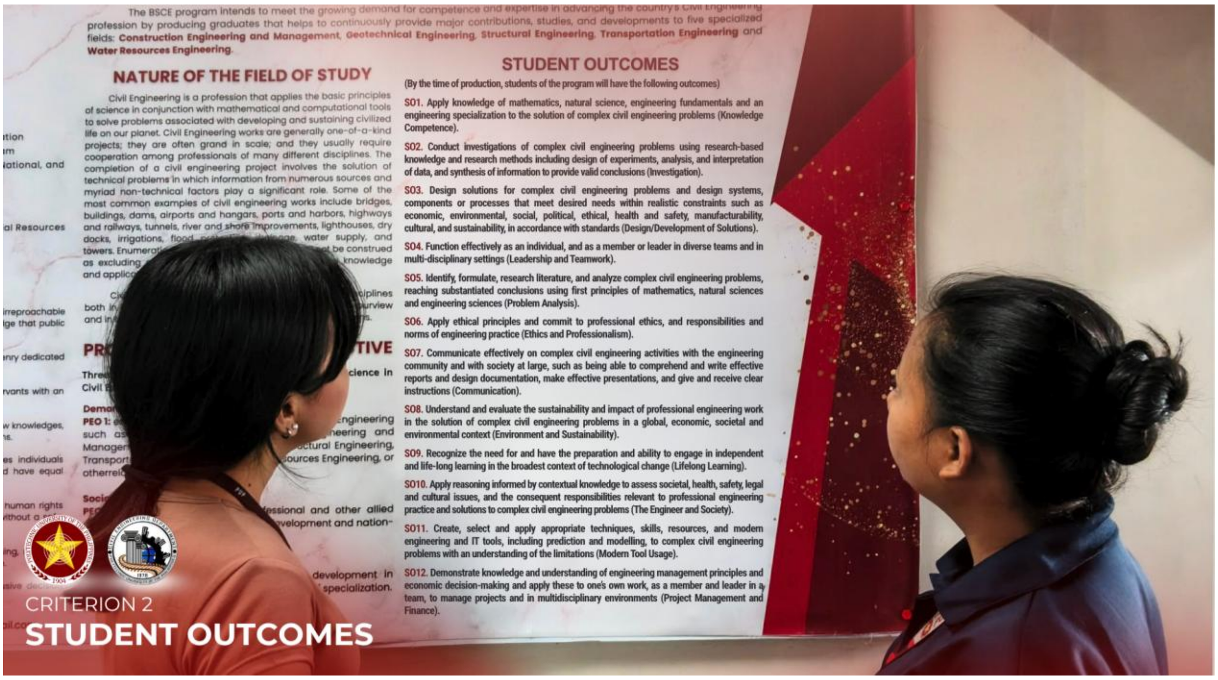

Figure 2-18. *Civil Engineering Department Bulletin Board located at CEA 2nd Floor showcasing key information, guidelines, essential updates for students and faculty and Student Outcomes.*

Republic of the Philippines
POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
Office of the Vice President for Academic Affairs
College of Engineering
CIVIL ENGINEERING DEPARTMENT

Student Outcomes Map Addressed by the Course	By the time of graduation, the students of the program shall have the ability to: SO 1. Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization to the solution of complex civil engineering problems. SO 2. Conduct investigations of complex civil engineering problems using research-based knowledge and research methods including design of experiments, analysis, and interpretation of data, and synthesis of information to provide valid conclusions. SO 4. Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings. SO 5. Identify, formulate, research literature, and analyze complex civil engineering problems, reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences. SO 7. Communicate effectively on complex civil engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. SO 10. Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues, and the consequent responsibilities relevant to professional engineering practice and solutions to complex civil engineering problems. SO 11. Create, select and apply appropriate techniques, skills, resources, and modern engineering and IT tools, including prediction and modelling, to complex civil engineering problems with an understanding of the limitations.

CRITERION 2
STUDENT OUTCOMES

	CIEN 301	COURSE TITLE	GEOTECHNICAL ENGINEERING 1 (LEC)
CREDIT UNITS:	3 UNITS	COURSE PRE-REQUISITES / CO-REQUISITES:	CIEN 203 – Geology for Civil Engineers ENSC 019 – Mechanics of Deformable Bodies
COURSE DESCRIPTION:	The course covers topics related to soil mechanics. A progressive requirement of the course is the common terminology used in the field of geotechnical engineering, derivation and calculation of geotechnical parameters, analysis of these parameters and the analysis of soil behavior.		

Figure 2-19. *Student Outcomes Map for CIEN 301 (Geotechnical Engineering 1) OBTLP, outlining the alignment of course content with program learning outcomes of the Civil Engineering Department.*

Figure 2-20. *Salubong 2025: Freshmen Orientation held on November 8, 2025, at Bulwagang Balagtas, PUP Main, Sta. Mesa is an Orientation plus a Celebration of connection, enthusiasm, and new beginnings for students of BS Civil Engineering, Railway Engineering Technology, and Civil Engineering Technology.*

Figure 2-21. *Engr. Lorenzo Galicia, faculty member of the Civil Engineering Department and current QA/QC Coordinator, delivers an engaging talk on Student Outcomes during the Freshmen Orientation at Bulwagang Balagtas, PUP Main Campus, Sta. Mesa, Manila.*

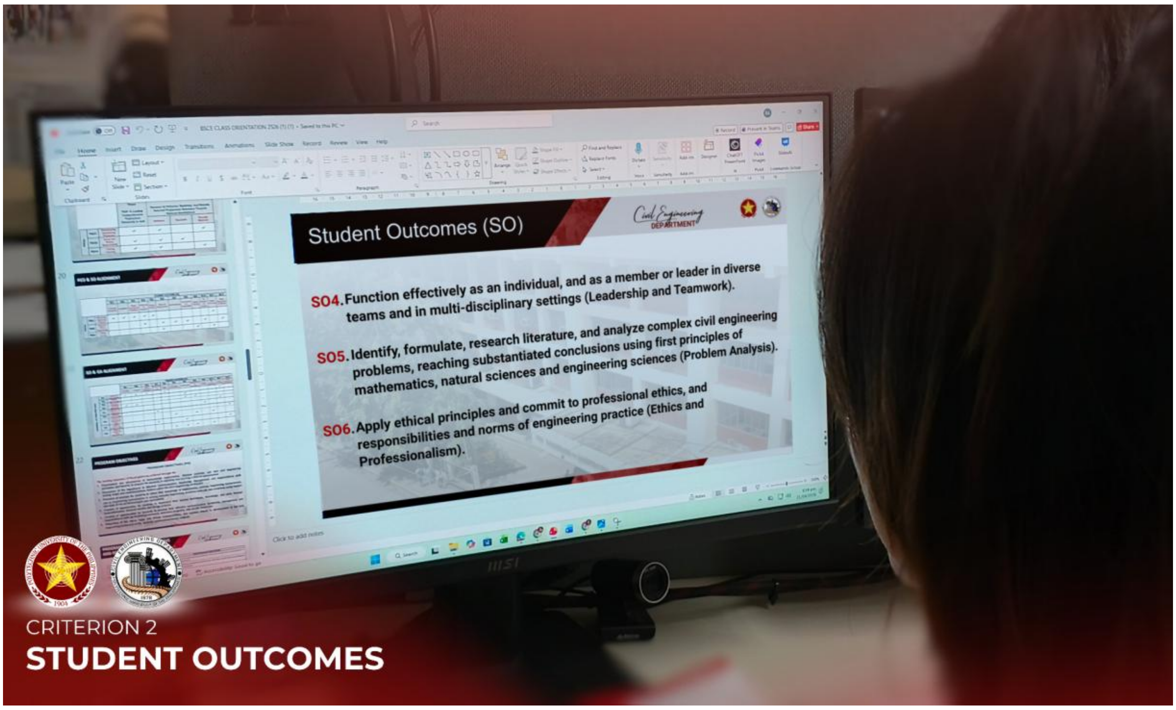

Figure 2-22. *The department provides a standardized PowerPoint presentation template for use during the first meeting of each class, incorporating a dedicated section on Student Outcomes to ensure consistent and aligned delivery across all courses.*

Figure 2-23. *Engr. Ramir M. Cruz emphasizes Student Outcomes at the start of each class, guiding students in understanding the alignment of course objectives and activities with the attainment of these outcomes.*

Through these multiple channels and activities, the department ensures that the Student Outcomes are effectively communicated, internalized, and aligned with the learning and assessment strategies of the Civil Engineering program as shown in Figure 2.5.6 below.

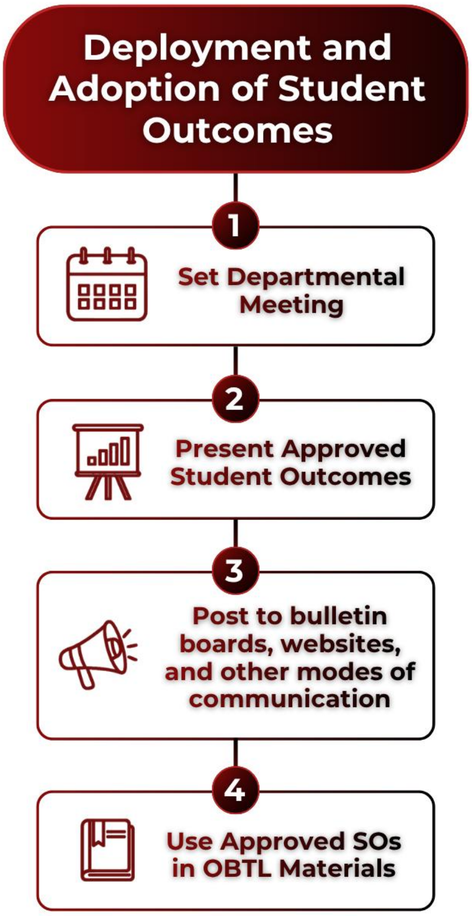

Figure 2-24. *This diagram illustrates the integration of Student Outcomes within the program's Outcomes-Based Education framework, mapping their alignment from national and international standards down to classroom implementation. It highlights the interconnected roles of stakeholder engagement, structural assessment methodologies, and the continuous quality improvement loop that ensures long-term graduate competence.*

The established SOs of the BSCE program serve as the cornerstone of the program's OBE framework. These outcomes are systematically aligned with national standards prescribed by CHED, accreditation requirements of the PTC, and international benchmarks such as the Washington Accord, thereby ensuring both local relevance and global comparability.

Through a structured process of formulation, deployment, assessment, and continuous improvement, the program ensures that all Student Outcomes remain measurable, attainable, and responsive to the evolving demands of the civil engineering profession. The integration of clearly defined performance indicators, consistent assessment methodologies, and active stakeholder engagement reinforces the reliability and validity of outcome attainment.

Furthermore, the strong linkage between SOs and PEOs guarantees that graduates are not only equipped with fundamental engineering competencies but are also prepared for professional practice, lifelong learning, and meaningful contributions to society. The deployment mechanisms, which include curricular integration, faculty alignment, and student orientation, ensure that these outcomes are effectively communicated and internalized across all levels of the program.

The implementation of a robust CQI system demonstrates the program's commitment to excellence. By utilizing assessment data to drive decision-making, close identified gaps, and refine both curriculum and outcomes, the program sustains a dynamic and responsive educational environment. This continuous feedback loop ensures that improvements are evidence-based, documented, and aligned with institutional and accreditation standards.

In summary, the BSCE program has established a comprehensive, systematic, and sustainable approach to the development and attainment of Student Outcomes. This approach ensures that graduates possess the necessary competencies to address complex engineering challenges, uphold professional and ethical standards, and contribute effectively to national development and global engineering practice.

2.6 ATTACHMENTS

- Attachment C2-01 – Board Resolution No. 3587
- Attachment C2-02 – Board Referendum No. 162 s 2024
- Attachment C2-03 – Sample Syllabus
- Attachment C2-04 – Endorsement for the BOR Approval of College of Engineering Programs' PEOs
- Attachment C2-05 – GIAB Minutes of Meeting dated April 8, 2024
- Attachment C2-06 – GIAB Minutes of Meeting dated April 11, 2024
- Attachment C2-07 – GIAB Minutes of Meeting dated May 30, 2025
- Attachment C2-08 – GIAB Minutes of Meeting dated March 16, 2026
- Attachment C2-09 – UCEC Minutes of Meeting dated July 19, 2024
- Attachment C2-10 – Approved Revised 2022 Curriculum for the Bachelor of Science in Civil Engineering (BSCE) Program
    - Attachment C2-10.1 – Approved Revised 2022 Curriculum for the Bachelor of Science in Civil Engineering (BSCE) Program
    - Attachment C2-10.2 – CHED Memorandum Order No 92 series of 2017 Policies, Standards and Guidelines for Bachelor of Science in Civil Engineering (BSCE) Program
    - Attachment C2-10.3 – Memoranda and Minutes of Meetings for BSCE Curriculum Revision
    - Attachment C2-10.4 – BSCE Graduates Online Tracer Study
    - Attachment C2-10.5 – Special Order No 0955 for the designation of the Curriculum Development Committee (CDC) of the BS Civil Engineering Program
    - Attachment C2-10.6 – Special Order No. 1761 for the designation of the College of Engineering Bachelor of Science in Civil Engineering Program Government Industry Advisory Board (GIAB)
    - Attachment C2-10.7 – BSCE Program Executive Summary
    - Attachment C2-10.8 – Endorsement Form
    - Attachment C2-10.9 – Equivalency of Courses Based on Elective Course, Accreditation of Ladderized Program Courses into the Bachelor's Degree Program
    - Attachment C2-10.10 – Minutes of the UCEC Meeting
- Attachment C2-11 – PowerPoint Presentation of Class Orientation
- Attachment C2-12 – Curriculum Development/Revision Guidelines in 2026
- Attachment C2-13 – 2018-2022 CQI for Student Outcomes Assessment